Lab 11 – Worksheet

Name:	ID:	Section:

Task a: vga_sync

Code: Design module `vga_sync`

Provide appropriately commented code for designed module, the code should contain meaningful variable naming.

**Add snippet or Copy paste the code written in the Editor window. Make sure the irrelevant area of the snip is cropped.*

Results (Waveforms)

**Add snip of relevant signals' waveforms. Make sure the irrelevant area of the snip is cropped. make sure to scale properly for visibility of all case.*

Comments

**Observation/Comments on the obtained results/working of code.*

Task b: Pixel Generation

Code: Design pixel_gen

Provide appropriately commented code for designed module and the testbench, the, code should contain meaningful variable naming.

**Add snippet or Copy paste the code written in the Editor window. Make sure the irrelevant area of the snip is cropped.*

Task c: Top-Level Module/VGA Controller

Create a Top-Level Module as in Figure 11. 7. This module will receive a 100Mhz Clock as input and generate 1-bit *h_sync* and *v_sync* signals and three 4-bit signals for Red, Green and Blue color.

Code: Design module & testbench

Provide appropriately commented code for designed module & its testbench, code should contain meaningful variable names.

**Add snippet or Copy paste the code written in the Editor window. Make sure the irrelevant area of the snip is cropped.*

Constraint File

Copy & paste your constraint file (.xdc) here:

Results

Implement[1] the VGA controller on BASYS-03 board and show output on VGA monitor. The display output is shown in Figure 11. 8. The VGA port connections are shown in Figure 11. 9.

Demonstrate your working VGA monitor to respective RA. Attach the picture of your output screen.

Comments

**Observation/Comments on the obtained results/working of code.*

Exercise

Code: Design module & testbench

Provide appropriately commented code for designed module & its testbench, code should contain meaningful variable naming.

**Add snippet or Copy paste the code written in the Editor window. Make sure the irrelevant area of the snip is cropped.*

Results (Waveforms)

Attach a clear picture of the Monitor output here

Comments

**Observation/Comments on the obtained results/working of code.*

Assessment Rubric
Lab 11
VGA Controller

Name:	Student ID:

Marks Distribution:

	Task No.	LR2 Code	LR5 Results
In - Lab	Task a	/20	/05
	Task b	-	-
	Task c	/10	/15
Exercise		/20	/15
Total: 100		/50	/35
CLO Mapping		CLO3	CLO3

Affective Domain Rubric		Points	CLO Mapped
AR 7	Report Submission and Viva	/15	CLO 4

CLO	Total Points	Points Obtained
3	85	
4	15	
Total	100	

For description of different levels of the mapped rubrics, please refer to the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

Lab Evaluation Assessment Rubrics

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3: Good Points 3	Level 4: Exemplary Points 4
LR1	**Circuit Layout**	Connections between circuit components are mostly wrong. Circuit layout is cluttered. Needs guidance to make correct equipment/ component connections.	Few of the connections made between circuit components are wrong. Circuit layout is not neat and clean as per standards.	Circuit layout and connections are correct but not all connections are neat and clean as per standards.	Neat, clean and correct connections of all the circuit components/ equipment are made as per standard circuit diagram.
LR3	**Troubleshooting**	Unable to identify the fault/minimal effort shown in troubleshooting.	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and takes necessary steps and actions to correct it.
LR5	**Results & Plots**	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR7	**Viva**	Response shows a complete lack of understanding of the assigned / completed task	Response shows shallow understanding of the assigned task	Response shows substantial understanding of the assigned task	Response shows complete understanding of the completed task. The student is able to explain all the related concepts.
LR10	**Analysis**	Results are not interpreted or are analyzed incorrectly.	Analysis of the obtained results is incomplete. Conclusions are not drawn.	Results are analyzed and concluded but are not well explained and justification is not provided with the help of reasoning.	Results are well analyzed supported with reasons. Good comparison is made between the theoretical and experimental results with correct and useful conclusion.
LR11	**Design**	Proposed design is unsubstantiated or	Justification for proposed design is	Proposed design is substantiated,	Proposed design is substantiated,

		does not satisfy most of given constraints. The breakdown of system into features is incomplete and still at lower resolution.	weak, and it only satisfies some constraints. The breakdown of system is missing some features and resolution is not appropriate at some places.	preferably through proper analysis, and satisfies most given constraints. The breakdown of system into features seems complete, but resolution is not appropriate at some places.	preferably through proper analysis, and satisfies all given constraints. The breakdown of system into features seems complete and at appropriate resolution, given students' level.